Soumia Amari

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Profile

Final year student at the Higher School Of Computer Science (ESI) of Sidi Bel Abbès, specializing in **Artificial Intelligence and Data Science**. Passionate about AI, I am currently preparing my Master's and State Engineer degrees in Computer Science.

Education

2022–2027: **Higher School Of Computer Science, Sidi Bel Abbès** — Engineering Cycle in Computer Science, Specialization: Artificial Intelligence and Data Science

2022: **B**accalaureate in Experimental Sciences, With Highest Honors (17.48/20)

Academic and Personal Projects

Meetly: Team project : Intelligent platform for business meetings in **Algerian dialect (Darja)**. Fine-tuning of **Whisper large-v3** with **LoRA/PEFT** for ASR, speaker diarization with **pyannote.audio**, and automatic summarization with an **LLM**. Interactive web application (FastAPI, React). Handling of Arabic/French code-switching, original dataset, audio augmentation and algorithmic cleaning. .

Multimodal Enterprise RAG: **RAG** agent answering questions on PDF documents with source citations (pages, scores). Parsing with **LlamaParse**, vector indexing in **Qdrant**, cross-encoder **reranking** and **Self-RAG** (relevance evaluation, query rewriting, rejection on low confidence) to reduce hallucinations. LLM via Groq, **LlamaIndex**, **Chainlit** chat interface, **Docker**.

Emotion Recognition: Team project: facial emotion recognition (7 classes) with a **CNN** (Keras/TensorFlow, Batch Normalization, Dropout, Adam) trained on the FER dataset (35,887 48×48 images). **Streamlit** web app. My role: design and development of the Deep Learning model.

Text Summarization: Team project: NLP web app for abstractive text summarization with a fine-tuned **BART** model. Input from PDF upload or direct text, summary history, statistics dashboard and PDF report export. **Streamlit**.

Autism Detection: Machine Learning project to identify autistic traits in children from behavioral data. **Stack:** Python, Scikit-learn, Pandas, NumPy. EDA and classification.

EduWorld: Online course platform (full-stack). **Backend:** Node.js (Express.js), MongoDB. **Frontend:** React, Tailwind, HTML, CSS. Secure authentication and RBAC.

Supp Hours: Web application for overtime management. **Backend:** Node.js (Express.js), MySQL. Secure authentication and RBAC.

MyVet: E-commerce project. Role: **UI/UX Designer** (high-fidelity mockups in Figma).

Technical Skills

ML & Deep Learning: **TensorFlow, Keras, PyTorch**, Scikit-learn, Pandas

ML Algorithms: Linear/Logistic Regression, Naïve Bayes, KNN, SVM, Decision Trees, Random Forest, Gradient Boosting, XGBoost; K-Means, Hierarchical Clustering, DBSCAN; PCA, Kernel PCA, SVD, NMF; cross-validation, regularization

Deep Learning Models: MLP, CNN (ResNet, MobileNet, EfficientNet), RNN, LSTM, GRU, Autoencoders, GAN, Transfer Learning; Adam optimizer, Batch Normalization

NLP & RAG: RNN, LSTM, BERT, BART, LlamaIndex; RAG, vector embeddings, reranking; tokenization, lemmatization, NER, POS tagging (HMM, Viterbi), Bag of Words, word embeddings, seq2seq, attention mechanisms, neural machine translation

Computer Vision: Image classification, object detection, image segmentation

MLOps: Flask, model deployment via REST API, MLflow, TensorFlow Serving, CI/CD

Prog. & Web: Python, Java, JavaScript, SQL; React, Node.js, Express.js, HTML/CSS, Tailwind, Streamlit

Databases: MySQL, PostgreSQL, MongoDB, Neo4j, Qdrant, Cassandra

Tools & Design: Docker, Git, Postman, Figma, Adobe Illustrator, Photoshop

Professional Experience

Sept. 2025: **National Transmission Directorate**, Tissemsilt — Practical internship. Observation of and contribution to equipment maintenance and network installation monitoring.

Extracurricular Activities

Member, External Relations Department — **Ingenuims Club**, ESI Sidi Bel Abbès

Member, Design Department — **Google Developer Student Club (GDSC)**, ESI Sidi Bel Abbès

Member, video editing sub-department — **Alphabit Club**, ESI Sidi Bel Abbès

Languages

Arabic (native) – French (TCF C1) – English (B2) – Spanish (beginner)

Certifications

Oct. 2026: Kaggle — Feature Engineering, Intermediate Machine Learning, Intro to Machine Learning, Computer Vision, Machine Learning Explainability